Lab Assignment – 3

JavaScript Programs | Deadline: 22 October 2026

Each program has a separate HTML file and JavaScript file. No CSS file is used.

1. BMI Calculator

BMI.html

```
<h2>BMI Calculator</h2>

Height (in meters):
<input type="number" id="h"><br><br>

Weight (in kg):
<input type="number" id="w"><br><br>

<button onclick="bmi()">Calculate</button>

<p id="result"></p>

<script src="BMI.js"></script>
```

BMI.js

```
function bmi() {
    let h = document.getElementById("h").value;
    let w = document.getElementById("w").value;

    if (h == "" || w == "" || h <= 0 || w <= 0) {
        document.getElementById("result").innerHTML = "Enter valid values";
        return;
    }

    let b = w / (h * h);
    let category;

    if (b < 18.5)
        category = "Underweight";
    else if (b < 25)
        category = "Normal";
    else
        category = "Overweight";

    document.getElementById("result").innerHTML =
        "BMI = " + b.toFixed(2) + "<br>" + category;
}
```

2. Signup Page

Signup.html

```
<h2>Signup Form</h2>

<form onsubmit="return signup()">
    Name: <input type="text" id="name" required><br><br>
    Email: <input type="email" id="email" required><br><br>
    Password: <input type="password" id="pass" required><br><br>
    Confirm Password: <input type="password" id="confirm" required><br><br>

    <button type="submit">Sign Up</button>
</form>

<p id="result"></p>

<script src="Signup.js"></script>
```

Signup.js

```
function signup() {
    let name = document.getElementById("name").value;
    let pass = document.getElementById("pass").value;
    let confirm = document.getElementById("confirm").value;

    if (pass.length < 8) {
        alert("Password must be at least 8 characters");
        return false;
    }

    if (pass != confirm) {
        alert("Password does not match");
        return false;
    }

    document.getElementById("result").innerHTML =
        "Signup Successful " + name;

    return false;
}
```

3. Change Color Theme

Theme.html

```
<h2>Change Color Theme</h2>

<button onclick="changeColor('white')">Light</button>
<button onclick="changeColor('black')">Dark</button>
<button onclick="changeColor('lightblue')">Blue</button>

<script src="Theme.js"></script>
```

Theme.js

```
function changeColor(color) {
    document.body.style.backgroundColor = color;

    if (color == "black")
        document.body.style.color = "white";
    else
        document.body.style.color = "black";
}
```

4. Digital Clock

Clock.html

```
<h2>Digital Clock</h2>

<h1 id="clock"></h1>

<script src="Clock.js"></script>
```

Clock.js

```
function showTime() {
    let time = new Date();

    document.getElementById("clock").innerHTML =
        time.toLocaleTimeString();
}

showTime();
setInterval(showTime, 1000);
```

5. To-Do List

Todo.html

```
<h2>To-Do List</h2>

<input type="text" id="task" placeholder="Enter task">
<button onclick="addTask()">Add</button>

<ul id="list"></ul>

<script src="Todo.js"></script>
```

Todo.js

```
function addTask() {
    let task = document.getElementById("task").value;

    if (task == "") {
        alert("Enter a task");
        return;
    }

    let li = document.createElement("li");
    li.innerHTML = task;

    li.onclick = function() {
        li.style.textDecoration = "line-through";
    };

    let btn = document.createElement("button");
    btn.innerHTML = "Delete";

    btn.onclick = function(event) {
        event.stopPropagation();
        li.remove();
    };

    li.appendChild(btn);
    document.getElementById("list").appendChild(li);

    document.getElementById("task").value = "";
}
```